Negotiated Autonomy

How Personal AI Learns Its Scope of Action in Spatial Work

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Abstract

Personal AI increasingly turns conversation into action, but understanding a goal does not establish permission to pursue every means. I propose that its scope of action should be negotiated through ongoing work. In spatial tasks, permission has geometry: people indicate where changes belong, inspect previews and authorise their commitment. I will build a desktop/XR collaborator whose permissions are visible and revisable, and whose adaptation learns from corrections without treating silence as consent. The research will explain how people grant and revise delegation, test spatial mechanisms for doing so, and develop measures of autonomy drift across repeated use.

1 Motivation

A user says, “Make this workspace suitable for collaboration, but keep the window workstation.” The AI improves the layout and also moves a monitor the user expected to remain untouched. The arrangement meets the broad goal yet exceeds the intended delegation. Later, the same user explicitly permits the AI to rearrange chairs while they are away. Both episodes concern how useful action becomes authorised.

My broader goal is a personal AI that communicates, perceives its context, reasons and executes. Here, personal includes learning how far a particular person wants to delegate as their working relationship develops. XR gives delegation a visible spatial form: a floating room model displays alternatives, a gesture marks an authorised region, and an object-linked label announces the next change. A palm gesture or spoken interruption returns control.

2 Related Work

Intent formation. Rambler [1], Orality [2] and IntentPrism [3] support expressing, developing and inspecting intentions. These approaches inform how a person articulates a request. My question concerns the subsequent delegation: which actions does that request authorise, on which objects, and for how long?

Systems that act. Reference-Based Manipulation formalises spatial references [4]; EchoLadder supports progressive scene design with preview, application and undo [5]. MUSE [6], SpatiaLLM [7] and LLMR [8] connect language to scene operations, while OmniActions predicts digital actions from multimodal input [9]. These systems offer foundations for execution and user control. The open question here is how users express and revise permissions that extend across sequences of spatial actions and repeated sessions.

Mixed initiative and adaptive automation. Horvitz treats initiative as a decision under uncertainty [10]; Parasuraman, Sheridan and Wickens distinguish automation types and levels [11]; Amershi et al. specify support for correction and adaptation [12]. Sensible Agent further addresses unobtrusive proactive AR assistance [13]. Adaptive autonomy is therefore established territory. The proposed contribution is an empirical account and spatial interface for negotiating object-specific permissions as work and familiarity change.

3 Research Questions

RQ1: How do people grant, withdraw and recalibrate an AI's scope of action while speaking and manipulating a scene?

RQ2: How should that scope be made visible, editable and interruptible, and how should correction, non-response and takeover inform adaptation?

RQ3, longitudinal: How do permissions change with repeated use, and which changes represent wanted delegation or unwanted drift?

4 System and Interaction Mechanisms

A language model proposes edits from speech and editor state; a permission check gates the tool that changes the scene. This connects communication, contextual perception and execution in one workflow.

4.1 Making scope visible

I will adapt Source/Anchor/Frame [4] to specify the spatial part of permission. For “move only the chairs on this side of the desk”, the desk supplies the Source, its edge supplies an Anchor, and the user's viewpoint supplies the Frame; chairs remain the action targets. A highlighted region shows inclusion and exclusion. A floating label adds the allowed operation, duration and execution mode. Geometry locates permission; these additional terms determine what is authorised. Timestamped records distinguish the scene when the user spoke from the scene when an action executes.

4.2 Degrees of commitment

Building on EchoLadder's previews and undo [5], the prototype separates suggestions, ghost previews, sandbox execution and commitment to the shared scene. Previews are discarded; sandbox changes reset; committed edits retain an undo history. Background completion requires a separate, time-limited grant because absence changes oversight, not reversibility. A fixed asset library and resettable rooms support these interactions.

4.3 Learning from corrections

A per-user estimator will rank execution modes using action type, prior approvals and confirmed corrections. I will initialise it from Study 1 episodes and update it with feedback during use. Unexplained silence remains ambiguous evidence. The policy selects a mode within an explicit permission ceiling; expanding object scope, operation or duration requires a new grant. Repeated acceptance supports a proposal to delegate further. A compact task record stores authorised actions, their scope and expiry, together with the evidence and revisions that produced them.

4.4 Interrupting and returning control

A spoken stop or palm gesture cancels pending edits in the indicated scope. Manual takeover pauses edits to the touched object while its meaning is clarified. The interface marks completed changes, offers undo and explains the handover. Cancelled work resumes only after a renewed instruction.

5 Methods and Evaluation

Study 1: Authorisation behaviour. Fourteen participants will redesign workspaces with a Wizard-of-Oz assistant. Scripted opportunities to delegate, retract and leave work running will be combined with spontaneous interaction. Recordings and stimulated recall will produce a taxonomy of authorisation acts, including disputed and ambiguous cases. Participants' explanations are essential evidence about scope.

Study 2: Comparing delegation policies. Thirty participants will complete counterbalanced tasks under fixed low autonomy, fixed high autonomy and adaptive boundaries. Low autonomy requests commitment for each edit; high autonomy executes within the current grant; adaptive boundaries adjust execution modes and propose scope revisions from confirmed feedback. All conditions share models, tools, displays and initial grants, accept explicit scope updates, and retain stop and undo. I will preregister unwanted actions and task completion as primary outcomes, estimate condition differences with participant and task effects, and report effect sizes and uncertainty.

Measure	Operational definition
Autonomy drift	Mismatch between executed actions and intended delegation, using pre-task statements and event review; delayed discovery coded separately.
Undo and regret	Reverted actions and later regret, distinguishing exploratory undo from unwanted intervention.
Interruptions and repair	Stops, takeovers and clarification requests counted separately; time and edits needed to recover intended work.
Trust calibration	Predicted scope adherence and task success compared with observed outcomes and reliance decisions.
Task completion	User-endorsed outcomes, elapsed time, latency and checking effort.

Formal permission violations will be audited separately from expectation mismatch, which also arises inside a broad grant. Evaluation will distinguish model claims, successful tool execution and achievement of the user's goal. Retrospective accounts will distinguish changed preferences from unwanted actions at the time; disputed cases remain separately coded.

Study 3: Moving boundaries. Twelve participants will use the prototype for six weeks on recurring design tasks, answering RQ3. Weekly reviews will compare recorded grants with desired delegation, tracing deliberate expansion, withdrawal and unnoticed drift.

Preliminary Work

I built Game Fellows, a bounded, rule-based game companion, and prepared a counterbalanced Solo/Fellow comparison with paired random seeds, event logs and a questionnaire including perceived control. Human-participant evaluation is planned and has not been completed. This work provides experience building observable assistance and instrumenting comparisons.

6 Contributions and Schedule

The thesis will contribute an empirical account of how delegation is negotiated, tested spatial mechanisms for revising it, and a method for measuring autonomy drift. Its central claim is that permission should remain an inspectable part of collaboration as the AI becomes more capable.

Period	Work and output
Year 1	Wizard-of-Oz study; authorisation taxonomy; desktop/XR prototype.
Year 2	Spatial permissions, adaptation and interruption mechanisms; technical evaluation.
Year 3	Controlled comparison; analysis of drift and task outcomes.
Year 4	Six-week deployment; synthesis and thesis writing.

Studies will obtain ethics approval and informed consent, minimise recorded data, and support withdrawal and deletion.

7 Fit with Can Liu

Reference-Based Manipulation supplies a language for locating permission [4]. EchoLadder's reversible suggestions provide a starting point for negotiating commitment [5]. Orality motivates capturing grants while people develop thoughts aloud [2]. These connections make the spatial interaction itself part of the research.

I bring architectural training and CAVE installation experience, bounded companion prototyping through Game Fellows, and error-analysis practice from an image-detection capstone. In the first six months, I propose building two spatial permission interfaces and running the Wizard-of-Oz study to identify how users grant and revise delegation before implementing an adaptive policy.

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